



EPIC Readiness – Center Technical Requirements

Center Local Infrastructure and Workstation Components

Network

- WAN:
 - Connectivity: Redundant dedicated connections per site with DIA (Primary) and a regular Internet (Secondary) from different ISPs with an SD-WAN solution for auto failover.
 - SD-WAN Devices: Physical and Virtual redundancy with auto failover is required for all sites with more than 20 users.
 - Suggested SD-WAN Technologies:
 - Fortinet:
 - FG/FWF-40F
 - FG/FWF-60F
 - FG-90G
 - FG-120G/FG-200F
 - Bandwidth:
 - Recommended: At least 1Mbps x User. Ex: For a Center with 500 users = 500 Mbps
 - Min dedicated to Epic: 40 kb/s x Hyper Space Session (HSS). HSS = Concurrent Users x 2.
 - Min Total for any site: 50 Mbps.
- LAN:
 - Network Devices covered under vendor support with 4-hour replacement.
 - Network Devices with up-to-date software OS/Firmware.
 - Managed Business Grade devices. (Switches, Routers, Firewalls, etc.)
 - Gigabit Ethernet capable devices (Switches, Routers, Firewalls, etc.)
- WLAN:
 - 802.11 wireless n/ac/WiFi6 or newer
 - Network heatmaps for optimal Access Point (AP) placement and wireless coverage
 - Device configuration following industry standards and best practices.
 - Dual Band devices (2.4/5 GHz)
 - Centralized management
- General Design and Configuration:
 - Network design and device configuration following industry standards and best practices.
 - Network segmentation to improve performance, monitoring, and security.
 - Proper IT room cooling and cable management.
 - Redundant Uninterruptible Power Supply (UPS)
 - Conduct exhaustive network assessment and remediation.



Business Continuity Access (BCA)

- Workstation(s) to install a read only app to print downtime reports offline.
- Epic recommends 1 workstation per Department.
- Dedicated local printer connected with a USB cable.
- UPS to provide power for at least 30 min.
- EPIC recommends using an existing workstation.
- Workstation(s) needs to be in a separate segmented network
- Test workstation(s) twice a year

Workstations

- Client Workstation
 - Operating System:
 - Windows 11 (v23H2, v24H2) 64-bit (recommended)
 - Processor:
 - 13th,14th Gen or Series 1, or Series 2 Intel Core processor with a minimum of 4 performance cores @ 4.2 GHz max turbo frequency (or greater)
 - AMD Ryzen 5 or 7 7000 or 8000 series processor with a minimum of 4 physical cores @ 4.2 GHz max boost frequency (or greater)
 - Memory: 16GB+
 - Hard Drive: SSD
 - Video Device: DirectX 10.0 or higher supported
 - Display:
 - 23" widescreen for 1920 x 1080 resolution and 32-bit color
 - 24" widescreen for 1920 x 1200 resolution and 32-bit color
 - Network: Gigabit (10/100/1000) Network Interface Card or equivalent

Notes:

- **All Provider workstations need to have a webcam 1080p or higher compatible with Windows.**
- **Epic requires workstations in all patient rooms using an autologin workflow.**
- Canto Tablet
 - Operating System: Apple iOS
 - Device Type: Apple iPad
- Haiku Handheld Device
 - Operating System: Android OS or Apple iOS
 - Device Type: Android-based device, or Apple iPhone, iPad, or iPod Touch
- Welcome Kiosk
 - Operating System: Windows 11 (v23H2, v24H2) 64-bit (recommended)
 - Processor Type: Intel Core i5-6500 processor (launched Q3 2015) or better
 - Minimum Memory: 8 GB+ (16GB recommended)
 - Video Device: DirectX 10.0 or higher supported
 - Disk: 7200 RPM access speed or better
 - Monitor: 17" or larger, touch screen
 - Display: 1024x768 resolution, 32-bit color
 - NIC: Gigabit (10/100/1000) or equivalent
 - Other Hardware (Optional):
 - Barcode scanner
 - Camera
 - Magnetic card reader



- PIN pad
- Signature pad
- Welcome Tablet PC
 - Operating System: Microsoft Windows 11
 - Processor Type: Intel Atom Z3740 processor (launched Q3 2013) or better
 - Minimum Memory: 4 GB
 - Monitor: 10" or larger multi-touch capable
 - Display: 1920x1080 resolution or 1900x1200 resolution, 32-bit color
 - NIC: 802.11 wireless b/g/n or equivalent
- MiPACS Local Servers
 - MiPACS Storage Server MiniPACS Minimum System Configuration:
 - > 10 Clients: Microsoft® Windows Server 2016, 2019 (64-bit), or 2022
 - < = 10 Clients: Microsoft® Windows 11 Professional (64-bit)
 - Microsoft® SQL Server Express Edition 2022
 - 2.4+ GHz Intel Xeon 6+ Core Server for more than 10 Clients
 - 2.4+ GHz Intel i7 4+ Core Workstation for less than 10 Clients
 - 16 GB RAM
 - 1 Gbps Network Adapter
 - 1TB+ storage in RAID 1 array
 - Microsoft® .NET Framework 3.5 and 4.7
 - UPS device (Redundant for Servers)
 - MiPACS Central Regional Servers (Recommended for cluster of sites on large Centers)
 - Operating System (OS): Microsoft® Windows Server 2019 (64-bit), or 2022
 - SQL Server: Microsoft® SQL Server Standard Edition 2019, or 2022
 - Processor (CPU): 8 to 16 virtual CPUs (vCPUs).
 - RAM: 32-64 GB or more, depending on the scale and performance requirements.
 - Features: Microsoft® .NET Framework 4.8 or higher (excluding .NET Framework 4.8.1)
 - Storage: at least 1-10 TB of storage space (scalable based on your requirements).
- Workstation Naming Convention
 - Workstations that will be accessing Epic will need to adhere to the following naming convention parameters.
 - Names must detail center, location, department, and a numerical identifier.
 - Names must not contain special characters, dashes, or underscores.
 - Example:
 - Center = HCN
 - Location = Doral
 - Department = Enterprise Services
 - Number = 4
 - Name of the workstation could be:
 - HCNDORES04
 - HCNDORALES04
 - It is recommended that computer names are kept at 14 characters or less.



Peripherals

- Patient Photo Cameras
 - Cameras must be compatible with/recognized by Windows
 - Logitech C615 **(HCN Recommended)**
 - Microsoft LifeCam Studio
- Credit Cards
 - EMV Devices (computer chip - enabled credit card and encrypt sensitive credit card number data)
 - The terminal must be supported by Center payment gateway and Epic third-party payment device API implemented
 - EMV payment terminal vendors
 - Verifone P200 – IPA software required **(HCN Recommended)**
 - PAX A 80 – Cloud Payments **(HCN Recommended)**
 - SRedkey – Back Office Billing **(HCN Recommended)**

E-Signature Pads

- Recommended devices based on Epic's requirements:
 - Windows compatible touch devices
 - **NOTE: There** is a maximum allowance of two Topaz models across all centers. If your Center already has Topaz devices, please notify HCN as soon as possible with the model number. If your existing devices are compatible, this would avoid purchasing additional devices unnecessarily.
- Recommended Models
 - Topaz
 - Topaz T-S461-HSB-R or Topaz T-LBK755-BHSB-R **(HCN Recommended)**
 - Topaz B, BSB, and BBSB devices will use USB redirection in virtualized environments such as Citrix Virtual Apps.
 - Topaz HSB and HSX devices use USB redirection in virtualized environments such as Citrix Virtual Apps
 - Scriptel
 - Scriptel SlimShield LCD ST1580 ProScript **(HCN Recommended)**
 - Scriptel devices use USB redirection in virtualized environments such as Citrix Virtual Apps
 - Scriptel has additional software that can help connect the physical pad to the Scriptel plugin in cases where USB redirection is not consistently working.
- Barcode Scanners:
 - Recommended barcode scanners based on Epic's requirements:
 - Datalogic
 - Datalogic Gryphon GD44XX & GD45XX **(HCN Recommended)**
 - Datalogic Gryphon GM4500 (wireless)
- Biometrics:
 - Recommended devices based on Epic's requirements:
 - PatientSecure (by Imprivata)



- RightPatient (by M2SYS)
- Printers and Scanners
 - Laser/Multi-Function Printers
 - Business/enterprise class laser printing device
 - Supports and sustains the printing volumes the Center business needs require
 - Manufacturer - supported Windows print driver that works with your EPS servers' operating system
 - Compatible with PCL5 or newer

Note: For shared workstations and some services, printers and scanners need to be added locally

- Kiosk Printers
 - If print from Welcome module, printer must support 8.5 inches paper width
 - Thermal printer designed recommended
 - Recommended printers based on Epic's requirements:
 - Custom Engineering KPM216H thermal printer (USB interface)
 - Custom Engineering KPM216HII thermal printer (USB interface)
 - Practical Automation Model ITK 38 (USB interface)
- Kiosk Scanners (For those using the Welcome platform)
 - Must be TWAIN compatible
 - Recommended scanners based on Epic's requirements:
 - Ambir DS490i (8.5" full size)
 - Ambir DS490ix (8.5" full size)
 - Ambir DS687 (business cards)
 - Docketport 687 (business card)
 - Inuvio Ecoscan i6d (business card)
 - Plustek MobileOffice D620 (business card)
- Document Scanners
 - Must be TWAIN or WAI 2.0 Compatible
 - Fujitsu - fi8xxx (**HCN Recommended**) earlier fi models can be used.
- Label Printers Supported
 - Label Printers:
 - Dymo LabelWriter 450, 450 Turbo & 550
 - Zebra LP2824 Plus
 - Label Size: 2 1/4 x 1 1/4
- Print Drivers **NOT Recommended**
 - Canon Generic Driver. All Versions
 - Star TSP800II (TSP 847II) v2.1.0.0
 - ZDesigner v5
 - Brother Mono UPD (PCL) v1.10.0.0
 - Canon Generic Driver PCL6 v2.2.21.5, v3.2.0.0
 - HP UPD PCL6 v5.1-5.7.v5.9-6.6.0, v6.7
 - Konica Minolta UPD PCL5 v3.1
 - Kyocera KX Driver v6.0.27.26
 - Lexmark UPD v2.2.3-2.6, v2.8
 - Ricoh UPD PCL6 v3.12, v3.13, v4.5
 - Sharp Universal Driver PCL 6 v2.1.20.18



- Xerox GPD PCL6 v3.4.377.6.0
- Scanning into OnBase
 - Base-line requirements:
 - Driver Compatibility: Scanners should utilize either TWAIN, KOFAX, or ISIS scan drivers.
 - Speed:
 - High-Volume: 90ppm or greater (simplex)
 - Departmental: 60ppm or greater (simplex)
 - Registration Desk: 25ppm or greater (simplex)
 - ADF Capacity:
 - High-Volume: 200 pages or greater
 - Departmental: 80 pages or greater
 - Registration Desk: 35 pages or greater
 - Resolution: Up to 600dpi
 - Interface: USB 2.0/3.0; Ultra SCSI
 - Additional Capabilities: Automatic page size detection, Duplex, De-skew, Automatic orientation, Blank page skip, Error diffusion, Dither, Barcode, 2D Barcode.
 - High-Volume / Production:
 - Good:
 - Ricoh-FUJITSU Document Scanner fi-8820
 - Kodak Alaris Document Scanner i4250
 - Better:
 - Ricoh-FUJITSU Document Scanner fi-8930
 - Kodak Alaris Document Scanner i5650S
 - Best:
 - Ricoh-FUJITSU Document Scanner fi-8950
 - Kodak Alaris Document Scanner i5850S
 - Departmental:
 - Good:
 - Ricoh-FUJITSU Document Scanner fi-8150
 - Kodak Alaris Document Scanner i1190E
 - Better:
 - Ricoh-FUJITSU Document Scanner fi-8270 (flatbed)
 - Kodak Alaris Document Scanner i2820
 - Best:
 - Ricoh-FUJITSU Document Scanner fi-8190
 - Kodak Alaris Document Scanner i2900
 - Registration Desk/Nurse's station:
 - Good:
 - Ricoh-FUJITSU Document Scanner fi-8190
 - Kodak Alaris Document Scanner E1025
 - Better:
 - Ricoh-FUJITSU Document Scanner fi-8170
 - Kodak Alaris Document Scanner E1035